PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT COUNCIL
FORM FOUR PRE MOCK EXAMINATION
PHYSICS 1

Instructions

1. This paper consists of sections A,B and C
2. Answer all questions in sections A and B,and one question from section C.
3. Cellular phones and any unauthorized materials are no allowed in the examination room.
4. Non programmable calculators may be used
5. Write your examination number on every page of your answer sheet(s)
6. Where necessary the following constants may be used.
    i. Acceleration due to gravity,g=10m/s^2
    ii. Density of water =1000kg/m^3
    iii. Pie,=3.14

SECTION A(15Marks)

1. For each of the following items (i)-(x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided
   i. A body is said to be in equilibrium if
      A. Moves with the same speed
      B. The upward and downward force are equal
      C. Net force acting is zero
      D. Its centre of gravity is low
      E. Its centre of gravity is high
   ii. Presence of charges on a body can be detected by using
      A. Electroscope
      B. Gold leaf
      C. Capacitor
      D. Resistor
      E. Battery
   iii. The temperature of a certain liquid is measured to be 300K.what will be its temperature in degree centigrade?
      A. 30˚C
      B. 100˚C
      C. 57˚C
      D. 37˚C
      E. 27˚C
   iv. Is the ratio of sin i/sin r
      A. Critical angle
      B. Snells law
      C. Refractive index
      D. Law of refraction
      E. Law of reflection
   v. Alpha particles are
      A. Radioactive materials
      B. Parent nucleus
      C. Electrons
      D. Helium nucleus
      E. Electromagnetic waves
   vi. A force which temporary reduce the weight of the body immersed in a liquid is
      A. Gravity force
      B. Magnetic force

C. Weak nuclear force
D. Upthrust
E. Friction

vii. Water strides and mosquito can walk on the water surface.this is due to
A. Osmosis
B. Adhesion
C. Surface tension
D. Capillarity
E. Cohesion

viii. To pieces of metal are placed together.the direction of heat flow between them will depend on the
A. Direction of wind
B. Temperature difference between two metals
C. Length difference of two metals
D. Mass difference of two metals
E. Volume difference

ix. Why an atom is electrically neutral?
A. Has equal number of electrons
B. Has equal number of protons and neutrons
C. Has equal number of protons and electrons
D. Has equal number of protons
E. Has equal number of neutrons

x. The following is not a form of energy
A. Joule
B. Nuclear
C. Heat
D. Light
E. Chemical

2. Match the items in list A with the corresponding responses in list B by writing the letter of the correct response beside the item number in the answer sheet.

LIST A	**LIST B**
(i)the distance between principal focus and pole of the mirror (ii)the line joining the centre of curvature to the pale of the mirror (iii)the bending of light rays on passing through different media (iv)formed infront of a curved mirror	A.refraction of light B.Radius of curvative C.Principal focus D.Real image E.centre of curvative F.diffaction of light G.focal length H.Principal axis I.Virtual image J.Real image

SECTION B (60 Marks)

Answer all questions in this section.

3. (A)state three applications of density in daily life

(B) Briefly explain the following condititions

(i) Ice floats on water

(ii) a person can swim in sea water easily than in a fresh water

(C) An alloy contains 30% by mass of lead(density=11.4g/cm^3) and 70% by volume of tin (density=7.3g/cm^3 find the density of an alloy

4. (A)define the following apparatus
   I. Measuring cylinder
   II. Stop watch
   III. Thermometer
   IV. Galvanometer

(B) What should you do in the laboratory in the event of the following situations?

(i)your partner is cut cut by glass during experiment

(ii)you have some biscuits and your friend want to eat them

(iii) Your class leader wants to lock the laboratory door during experiment

(C) State three importance of first Aid

5. (A)During thunderstorm, the time between the flash of light and thunder is 10 seconds.explain this condition.

(B) Why bubbles of air increases in volume as it raises from the bottom of a pond of water to the surface

(C)explain two ways in which

(i)camera differ from human eye

(ii)camera is similar to human eye

6. (A)define the following terms as used in radio activity
   (i)nuclear fusion
   (ii)nuclear fission
   (iii)isotopes

(B)(i)atomic number and mass number are not the same,explain
(ii)the following symbol represents an atom of sodium.^{23}Na.finf its atomic number and mass number
(C) a random nucleus ^{222}Rn emits an alpha particle followed by a beta particle.what are the atomic number and mass number of the nuclei formed after emission.

7. (A)explain why
   I. A person doing high jumps prefer to land on sand instead of ground
   II. A mechanic chooses a long spanner to undo a tight nut.

(B)(i)define effort as used in a machine

(ii)a block and tackle pulley system consists of 5 pulleys.if the efficiency of the system is 80%.what load will be raised by an effort of 100N?

8. (A)the distance between successive crests of a sound wave is 6.2m and their wave speed is 340 m/s.calculate
   I. The wave length of sound wave
   II. The frequency of sound wave
   III. The period of sound wave

(B)the diagram below shows the electromagnetic apectrum.region D represesnts visisble light.

A	B	C	D	E	F	G

increasing frequency

I. which region represents radiationcapable of promoting skin to produce vtamin D
II. Which region contain radiation used to detect fracture f bones
III. Whish region contain radiation used in photosynthesis process

(C) give the name and source of radiation helpsa our eyes to see.

SECTION C (45 MARKS)

Answer any two (2) questions from this section

9. (A)(i) state the principle of conservation of energy
   (ii) give two applications of principle of conservation of energy
   (B)the diagram below shows a pendulum bob swinging from side to side.name the point at which the bob has.
   I. All potential energy
   II. All kinetic energy
   III. Partial potential and kinetic energy
   IV. Maximum speed.

(C)(i)how much work done to first lift a 7kg object a distance of 2m and then hold it at that height for 10 seconds?

(D)(i)name the device used to connect electrical energy to light energy

(ii)name the cell used to convert solar energy to electrical energy

(iii)name source of wind energy.

10. (A) how an echo differs from reverberation.give two points.

(B)in determing the depth of an ocean an echo sounder produces ultrasonic sound. Give two reasons why ultrasonic sound is preffered for this purpose.

(C)explain briefly why sound produsedin hall,with many people is heard more clearly than when the hall has few people?

(D)(i)a girl standing 200m from the foot of a high wall claps her hands and the echo reaches,her 1.16 seconds later.calculate the velocity of sounds later.calculate the velocity of sound in air using this observation.

(ii)why solid material transfer sound faster than liquid

(iii) Name at least one property of waves.

11. (A)define the following terms

I. Hypocenter
II. Epicenter
III. Magma

(B)(i)what is global warning?

(ii) Name four gases that contribute to global warming and give one source of each

(C)(i) Give three solutions to minimize global warming

(ii) The interior earth structure is divided into three zones.Name them.